

T.M.K.G THENNAKOON

SOFTWARE ENGINEERING INTERN

✉ kaveeshagimhan20@gmail.com

🏠 "Sanjeewani", Kudirippuwa, Galmuruwa.

☎ +94 776936850 🔄 GitHub 🌐 LinkedIn 📁 Portfolio



PROFILE

I'm Final-year undergraduate in Software Engineering with strong full-stack web development, embedded systems, and object-oriented programming expertise. Proficient in Java, Python, C++, Node.js, Express.js, and React, with hands-on experience in developing responsive web applications and API-based services. Experienced in both relational and NoSQL databases, including MySQL and MongoDB. Hands-on experience in Arduino programming and Embedded C, combining software and hardware in real-time applications. A diligent, fast learner passionate about creating efficient, user-friendly solutions. Actively seeking a software engineering internship to learn and contribute in a teamwork environment.

EDUCATIONAL QUALIFICATIONS

Bachelor of Science in Computer Engineering

2022 to Present

University of Jaffna, Sri Lanka

Relevant Coursework: Software Engineering, Object-Oriented Programming, Database Management Systems, Computer Networks & Security, Operating Systems, Computer Architecture, Embedded Systems Design, Data Structures & Algorithms, Machine Learning, Deep Learning, Intelligent Systems, Artificial Intelligence, Advanced Networking.

General Certificate of Education Advanced Level Examination

2018 to 2020

Dhammissara National Collage, Nattandiya

Results: A – Combined Maths B – Physics C – Chemistry

PROJECTS

01. BeautyBay (Cosmetic E-Commerce platform) 🔄

May 2025 to Present

Tech Stack:

ReactJS

MongoDB

ExpressJS

NodeJS

Tailwind CSS

Postman

- Designed and developed a full-featured e-commerce platform for cosmetic products using the MERN stack, supporting product browsing, cart management, and secure checkout.
- Implemented user authentication, product filtering, and dynamic UI interactions with React.js and Tailwind CSS for a modern, responsive interface.
- Used Postman for API testing to ensure backend functionality and reliability.
- Integrated Supabase for efficient and scalable image storage and retrieval across the platform.

02. Toyo AI Assistant (Chat & Translator Web App) 🔄

JULY 2025

Tech Stack:

Python

Streamlit

LangChain

Gemini API

- Developed a conversational AI web app featuring a chatbot powered by Gemini and a multilingual translator supporting 20+ languages via dynamic prompt chaining.
- Built using Streamlit for the frontend and LangChain for structured language model interaction and memory management.
- Integrated clean user input handling, maintained persistent chat history, and optimized for both local and cloud deployment.
- Secured API key usage with .env configuration, ensuring safe and scalable deployment practices.

03. Smart Water Management System (IOT Based Automated System)

Jan 2025 to May2025

Tech Stack: ReactJS Embedded C Arduino Firebase

- Designed and implemented a real-time water monitoring system using ESP32 and sensors to track temperature, TDS, water level, and inflow/outflow rates in water tank.
- Built a responsive React.js web dashboard with Firebase integration to display live data and interactive charts.
- Enabled remote motor and solenoid valve control through a cloud-connected UI, promoting efficient water usage and system automation.

04. Optimizing AI Algorithms for Low-Power and Edge Devices in Resource-Constrained Environments (Ongoing Research)

Jan 2025 to Present

Tech Stack: Python TFLite PyTorch ONNX

- Implemented a lightweight Transformer classifier to select the most suitable AI model based on input data type and device specifications.
- Designed a model ID system and model library with metadata such as RAM usage, power consumption, and performance metrics.
- Created a synthetic dataset including features like input size, RAM, power, accuracy, and type.
- Focused on AI deployment in resource-constrained environments like edge systems.

05. ConnectNow - Real-Time Video Conferencing App

Jan 2025 to May2025

Tech Stack: WebRTC Socket.IO JavaScript NodeJS Railway

- Developed a browser-based video conferencing platform supporting real-time multi-user communication.
- Implemented features like screen sharing, in-call messaging, emoji reactions, and recording functionality.
- Designed an interactive and responsive interface for seamless user experience during video calls.

SKILLS

- Programming Languages** | Java, C++, C, Python, PHP, HTML, CSS, JavaScript, MATLAB
- Frameworks** | React, NodeJS
- Databases** | MySQL, MongoDB
- Tools** | Google Colab, Jupyter Notebook, Git, Postman
- Embedded Systems** | Arduino, Embedded C
- Network and Security** | Wireshark, WebRTC
- Soft Skills** | Team Collaboration, Problem Solving, Self-Learning, Leadership

VOLUNTEERING EXPERIENCES

IESL Student Member

SEP 2023 to present

As a student member of the Institution of Engineers, Sri Lanka (IESL), I actively participate in professional development activities including technical seminars and engineering workshops. This involvement has allowed me to stay updated with current engineering trends and cultivate a strong foundation in professional ethics and leadership within the engineering community

REFERENCES

Dr. (Mrs.) J. Jananie

Senior Lecturer Grade II
Department of Computer Engineering,
Faculty of Engineering,
University of Jaffna
jananie@eng.jfn.ac.lk
+94 704904957

ENG.(Mr.) Y. Pirunthapan

Lecturer
Department of Computer Engineering,
Faculty of Engineering,
University of Jaffna
pirunthapany@eng.jfn.ac.lk
+94 776962062